

Project Design Phase Solution Architecture

Date	15 February 2025
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Overview

The solution architecture bridges the gap between the business problem (manual, inconsistent credit card approvals) and the technology solution (AI-powered prediction system). It defines how data flows, how components interact, and how the system ensures scalability, security, and efficiency.

Goals

- Identify the best technology stack to automate credit card approval decisions.
- Define the structure and behavior of the system for stakeholders.
- Specify features, development phases, and integration points.
- Ensure the solution is secure, scalable, and easy to maintain.

Architecture Description

The architecture consists of five layers:

Layer	Description	Technologies
1. User Interface Layer	Web portal for applicants and admin dashboard for bank staff.	HTML, CSS, React.js
2. Application Layer	Handles registration, login, and data validation.	Python (Flask Framework)
3. Machine Learning Layer	Predicts approval/rejection using trained models.	Scikit-learn, TensorFlow
4. Database Layer	Stores applicant data, model results, and logs.	MySQL / IBM DB2
5. Cloud Infrastructure Layer	Hosts the application and ensures scalability.	AWS / IBM Cloud / Kubernetes

Data Flow

1. Applicant submits data through the web interface.
2. Backend validates and stores data in the database.
3. ML model retrieves data and predicts approval or rejection.
4. Decision is sent back to the applicant dashboard.

5. Admin monitors model performance and system metrics.

